Practice Problem Sheet - 4
Transform Calculus
(MA20202)

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Q1. Solve $(D-2)x - (D+1)y = 6e^{3t}$

$(2D-3)x + (D-3)y = 6e^{3t}$ $\quad x(0)=3, y(0)=0$

Ans: $x = (1+2t)e^{t} + 2e^{3t}, \; y = (1-t)e^{t} - e^{3t}$

Q2. Solve $(D^2-1)x + 5Dy = t$ $\quad x(0)=y(0)=0$

$-2Dx + (D^2-4)y = -2$ $\quad x'(0)=y'(0)=0$

Ans: $x = -t + 5\sin t - 2\sin 2t, \; y = 1 - 2\cos t + \cos 2t$

Q3. Solve $\frac{dx}{dt} + \frac{dy}{dt} = t$ $\quad x(0)=y(0)=0$

$\frac{d^2x}{dt^2} - y = e^{-t}$ $\quad \left.\frac{dx}{dt}\right|_{t=0} = 0$

Ans: $x = \frac{t^2}{2} - 1 + \frac{1}{2}(e^{-t} + \cos t + \sin t), \; y = 1 - \frac{1}{2}(e^{-t} + \cos t + \sin t)$

Q4. Solve $F(t) = e^{-t} - 2\int_0^t F(u)\cos(t-u)\,du$

Ans: $F(t) = e^{-t}(1-t)^2$

Q5. Solve $\int_0^t F(u)F(t-u)\,du = 16\sin 4t$

Ans: $F(t) = \pm 8 J_0(4t)$

****** The End *******